

MOS Devices & Basic Circuits

Condensed Key-Points Study Notes

From Fundamental Physics to Advanced Architectures

With Hyperlink Resource References

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1. Solid-State Physics of the MOS Structure

NMOS Stack: Gate electrode → Oxide (SiO_2) → p-type semiconductor substrate.

- Gate electrode: historically Al; now **heavily doped polysilicon** (tunable work function, thermally resilient).
- Oxide (SiO_2): isolates gate from substrate — forms a **parallel-plate capacitor** for electrostatic channel control.
- Substrate: pure silicon is highly resistive; must be doped to become useful.

Doping Types:

- **n-type** — pentavalent dopants (P, As, Sb): 5th valence electron breaks free → electrons are majority carriers.
- **p-type** — trivalent dopants (B): introduces holes → holes are majority carriers.

NMOS Terminal Conventions:

- Source = common reference. V_{GS} = gate-to-source voltage (input). V_{DS} = drain-to-source voltage (output).
- Bulk (body) tied to most negative potential → keeps p-n junctions **reverse-biased**, preventing parasitic conduction.

Resource:

→ [MOS & MOSFET Review — UC Santa Barbara ECE225](#)

2. Electrostatics & the MOS Capacitor

Three electrostatic regimes as gate voltage is swept:

Regime	Gate Bias	Surface Condition
Accumulation	Negative (vs V_{fb})	Holes pile at interface — no lateral conduction.
Depletion	Small positive	Holes repelled; immobile -ve acceptor ions remain.
Strong Inversion	$V_{GS} \geq V_{th}$	Electrons drawn to surface; n-type channel bridges S→D.

Flat-Band Voltage (V_{fb}):

$$V_{fb} = \Phi_M - \Phi_S - Q_{ox}/C_{ox}$$

Applied to cancel work-function difference and trapped oxide charges so no band bending occurs.

Resource:

→ [n-channel MOSFET Electrostatics — TU Graz](#)

3. Threshold Voltage & Body Effect

Threshold Voltage (V_{th}): Minimum V_{GS} to achieve strong inversion.

- Sensitive to: substrate bias, temperature, process variation (Random Dopant Fluctuation).
- Thermal coefficient: typically **-2 to -4 mV/K** → V_{th} drops as device heats up, increasing leakage.

Body Effect (Shichman-Hodges Model):

$$V_{TN} = V_{TO} + \gamma (\sqrt{|V_{SB} + 2\phi_F|} - \sqrt{|2\phi_F|})$$

- $V_{SB} > 0$ widens the depletion region → gate needs higher voltage → V_{th} increases.
- $\gamma = (t_{ox}/\epsilon_{ox}) \sqrt{2q\epsilon_{Si}N_A}$ — body-effect parameter.
- Minimise γ by thinning t_{ox} , but ultra-thin oxides cause Fowler-Nordheim / direct tunneling leakage.

Fermi Potential:

$$\phi_F = (kT/q) \ln(N_A/n_i)$$

Resource:

→ [Threshold Voltage — Wikipedia](#)

4. Gradual Channel Approximation & Current Equations

GCA assumes vertical E-field $\gg$ lateral E-field → reduces 2D electrostatics to 1D current flow along channel.

Linear (Triode) Region — $V_{DS} < V_{GS} - V_{th}$:

$$I_D = \mu_n C_{ox} (W/L) [(V_{GS} - V_{th})V_{DS} - \frac{1}{2}V_{DS}^2]$$

Deep triode ($V_{DS} \ll V_{GS} - V_{th}$): transistor acts as a **voltage-controlled linear resistor**.

Saturation Region — $V_{DS} \geq V_{GS} - V_{th}$:

$$I_{D,sat} = \frac{1}{2} \mu_n C_{ox} (W/L) (V_{GS} - V_{th})^2$$

Channel pinch-off: $Q(L) \rightarrow 0$; current locked to $V_{GS} - V_{th}$. MOSFET = stable **voltage-controlled current source**.

Resources:

→ [Gradual Channel Approximation — MIT OCW](#)

→ [Purdue ECE 255 — MOSFET Equations](#)

5. NMOS vs. PMOS — Carrier Mobility Asymmetries

$\mu = q\tau/m^*$ — electron mobility (μ_n) $\gg$ hole mobility (μ_p) because electrons have lower effective mass in Si conduction band.

Metric	NMOS	PMOS
Mobility / Speed	High μ_n ; 2-2.5 $\times$ faster	Low μ_p ; slower
ON-Resistance	Lower R_{on}	Higher R_{on} ; more switching loss
Sizing for equal current	W_n (baseline)	$W_p \approx 2-2.5 \times W_n$
Leakage / Noise Immunity	Higher sub- V_{th} leakage	Lower leakage; better noise immunity
Control polarity	Conducts when $V_{GS} \gg 0$	Conducts when $V_{GS} \ll 0$

Key implication: Enlarging PMOS width to match NMOS drive strength increases gate capacitance $\rightarrow$ higher dynamic power & lower maximum clock frequency.

Resources:

→ [PMOS vs NMOS Basics — PatSnap Eureka](#)

→ [NMOS vs PMOS Comprehensive Comparison — Chipsmall](#)

6. Non-Ideal Behavior & Short-Channel Effects (SCEs)

As gate length shrinks past $\sim 1-2 \mu\text{m}$, GCA assumptions break down. Drain/source depletion fields compete with gate control.

Channel Length Modulation (CLM):

- Pinch-off point migrates toward source as V_{DS} increases $\rightarrow$ effective L decreases $\rightarrow I_D$ rises in saturation.
- Corrected equation: $I_{D,sat} = \frac{1}{2}\mu_n C_{ox} (W/L) (V_{GS} - V_{th})^2 (1 + \lambda V_{DS})$.
- Finite output resistance: $r_O \approx 1/\lambda I_D$ — degrades analog amplifier gain.

Velocity Saturation & Mobility Degradation:

- At high lateral E-fields, drift velocity saturates at v_{sat} (optical phonon scattering limit).
- $I_{D,sat}$ becomes **linear** in $(V_{GS} - V_{th})$ instead of square-law $\rightarrow$ reduced drive current vs. scaling prediction.

- Vertical E-field forces electrons against Si-SiO₂ interface → surface-roughness scattering → further mobility drop.

Drain-Induced Barrier Lowering (DIBL) & Threshold Roll-Off:

- DIBL: drain field lowers source-channel potential barrier → apparent V_{th} reduction at high V_{DS} .
- Threshold Roll-Off: shorter L → source/drain depletion 'share' gate's depletion charge → V_{th} decreases with L .

Subthreshold Swing, Punch-Through, GIDL:

- Subthreshold swing $S = (1 + (C_{dep} + C_s + C_d)/C_{ox}) \times 60 \text{ mV/dec}$. Ideal minimum at room temp: **60 mV/dec**.
- Short channels degrade S to 85–100 mV/dec → huge off-state leakage or higher V_{th} penalty.
- **Punch-through**: source/drain depletion regions merge → gate loses control → device unregulatable.
- **GIDL**: very negative V_{GD} → band bending triggers band-to-band tunneling → off-state leakage.

Resources:

- [Short-Channel Effects in MOSFETs — Bohrium Science](#)
- [Channel Length Modulation — Wikipedia](#)
- [Nanoscale SCE & FinFET/GAA Mitigation — All About Circuits](#)

7. Static CMOS Inverter — VTC & Noise Margins

Topology: PMOS (Pull-Up, PUN) from V_{DD} + NMOS (Pull-Down, PDN) to GND. Both gates share V_{IN} .

- $V_{IN} = 0 \rightarrow$ PMOS ON (linear), NMOS OFF $\rightarrow V_{OUT} = V_{DD}$.
- $V_{IN} = V_{DD} \rightarrow$ NMOS ON, PMOS OFF $\rightarrow V_{OUT} = 0$.
- **Zero static power** — no DC path between V_{DD} and GND at steady state (only subthreshold leakage).
- Output swings fully rail-to-rail: $V_{OH} = V_{DD}$, $V_{OL} = 0$.

Switching Threshold (V_M):

$$V_M = (V_{th,n} + \sqrt{(\beta_p/\beta_n)(V_{DD} + V_{th,p})}) / (1 + \sqrt{(\beta_p/\beta_n)})$$

For symmetric VTC: set $V_M = V_{DD}/2$ by matching $\beta_n = \beta_p \rightarrow W_p \approx 2W_n$.

Unity-Gain Points & Noise Margins:

- $V_{IL} = (3V_{DD} + 5V_{th,n})/8$ — max input still read as Logic 0.
- $V_{IH} = (5V_{DD} + 3V_{th,n})/8$ — min input read as Logic 1.

- $NM_L = V_{IL} - V_{OL} \mid NM_H = V_{OH} - V_{IH}$.
- Perfectly matched inverter: $NM_L = NM_H = V_{DD}/2$ — maximum noise immunity.

Resources:

- [CMOS Inverter — VTC, Noise Margins \(YouTube\)](#)
- [Noise Margins for the CMOS Inverter — UW ECE](#)

8. Combinational CMOS Logic & Equivalent Inverter Sizing

Complex gates: NMOS-only PDN + PMOS-only PUN. Topology governed by **De Morgan's duality** — series in PDN $\leftrightarrow$ parallel in PUN.

NAND vs NOR:

- **NAND:** NMOS in series (PDN); PMOS in parallel (PUN) — compact, preferred.
- **NOR:** NMOS in parallel (PDN); PMOS in series (PUN) — series PMOS must be doubled, already large $\rightarrow$ area bloat $\rightarrow$ avoid in high-speed datapaths.

Equivalent Inverter Sizing Rules:

- Series transistors: $R_{\text{eff}} = 2R$ (resistance doubles) $\rightarrow$ must widen by $2\times$ to maintain drive strength.
- Parallel transistors: $R_{\text{eff}} = R/2$ $\rightarrow$ no upsizing needed (each device independently drives the output).
- N-input NAND: NMOS widths scaled to $N \times W_n$; PMOS widths remain W_p .
- N-input NOR: PMOS widths scaled to $N \times W_p$ (already large) — exponential area cost.

Resources:

- [CMOS Gate Design — PDN/PUN Networks \(YouTube\)](#)
- [EEC 116 Lecture 5: CMOS Logic Sizing — UC Davis](#)
- [NTU Electronics — Chapter 15 CMOS Digital Logic](#)

9. Scaling Limits — FinFET & Gate-All-Around (GAA)

Planar MOSFET hit physical limits below 20 nm: gate on only one side could not suppress SCEs.

FinFET (commercialised at 22 nm node):

- Silicon channel etched into a 3D vertical 'fin'. Gate wraps **three sides** (tri-gate).
- Enhanced gate-channel coupling $\rightarrow$ suppressed DIBL, eradicated punch-through, near-ideal subthreshold swing.
- Limitation: fin base uncontacted $\rightarrow$ sub-fin leakage at ≤ 5 nm. Multiple fins side-by-side wastes area.

Gate-All-Around / Nanosheet (GAAFET):

- Channel completely severed from substrate; gate wraps **360° around** the channel.
- Vertically stacked rectangular silicon nanosheets → massive W_{eff} without increasing X-Y footprint.
- Superior electrostatic control; nanosheets → nanowires as EUV shrinks cross-sections further.
- GaN GAA nanosheet FETs: subthreshold swing at thermodynamic limit (60.50 mV/dec); THz cut-off frequencies → 6G candidates.

FinFET vs. GAA Nanosheet (3-5 nm node):

Metric	FinFET	GAA Nanosheet	Mechanism
DIBL	60-80 mV/V	30-45 mV/V	360° shielding blocks drain field from all angles; ~40-50% reduction.
Subthreshold Swing	75-85 mV/dec	65-70 mV/dec	Eliminated sub-fin leakage paths.
Off-State Leakage	Baseline (1×)	3.7-4× reduction	Steeper SS → near-ideal cutoff.
ON-State Drive Current	Baseline (1×)	~1.3× (30% boost)	Vertical stacking ↑ W_{eff} per footprint.

Resources:

- [MOSFET to FinFET to GAAFET: Evolution & Prospects — ResearchGate](#)
- [FinFETs Give Way to Gate-All-Around — Lam Research](#)
- [What Are GAA Transistors? — Synopsys Blog](#)
- [GaN Multi-Channel GAA Nanosheet FET — PMC Research](#)

Consolidated Reference Links

All primary sources used in these notes:

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2. [n-channel MOSFET — TU Graz](#)
3. [Threshold Voltage — Wikipedia](#)
4. [Short Channel MOS Transistor — Purdue Engineering](#)
5. [VLSI Design Lecture Notes — MMMUT](#)
6. [Gradual Channel Approximation — MIT OCW](#)
7. [ECE 255 — MOSFET Equations — Purdue](#)
8. [Channel Length Modulation — Wikipedia](#)
9. [PMOS vs NMOS Basics — PatSnap Eureka](#)
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12. [Mobility Degradation & Velocity Saturation — ResearchGate](#)
13. [Short-Channel Effects in MOSFETs — Bohrium](#)
14. [Nanoscale SCE & FinFET/GAA Mitigation — All About Circuits](#)
15. [MOSFET Channel-Length Modulation — All About Circuits](#)
16. [2nm GAAFET Nanosheet TCAD Analysis — U of Idaho](#)
17. [CMOS Inverter — VTC & Noise Margins \(YouTube\)](#)
18. [Noise Margins for CMOS Inverter — UW ECE](#)
19. [THE CMOS INVERTER — Textbook Chapter](#)
20. [NTU Electronics — Chapter 15 CMOS Digital Logic](#)
21. [CMOS Gate Design PDN/PUN \(YouTube\)](#)
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26. [What Is a Gate-All-Around Transistor? — ASML](#)
27. [GaN Multi-Channel GAA Nanosheet FET for THz — PMC](#)